

Lightweight Learning for Partial and Occluded Person Re-Identification

Arindam Sikdar[✉], *Member, IEEE* and Ananda S. Chowdhury[✉], *Senior Member, IEEE*

Abstract—Occluded and partial person re-identification (re-ID) problems have emerged as challenging research topics in the area of computer vision. Existing part-based models, with complex designs, fail to properly tackle these problems. The reasons for their failures are two-fold. First, individual body part appearances are not discriminative enough to distinguish between two closely appearing persons. Second, re-ID datasets typically lack detailed human body part annotations. To address these challenges, we present a lightweight yet accurate solution for partial person re-ID. Our proposed approach consists of two key components, namely, design of a lightweight unary–binary projective dictionary learning (UBDL) model, and construction of a similarity matrix for distilling knowledge from the deep omni-scale network (OSNet) to UBDL. The unary dictionary (UD) pair encodes patches horizontally, ignoring the viewpoints. The binary dictionary (BD) pairs, on the other hand, are learned between two views, giving more weight to less occluded vertical patches for improving the correspondence across the views. We formulate appropriate convex objective functions for unary and binary cases by incorporating the above knowledge similarity matrix. Closed form solutions are obtained for updating unary and BD components. Final matching scores are computed by fusing unary and binary matching scores with adaptive weighting of relevant cross-view patches. Extensive experiments and ablation studies on a number of occluded and partial re-ID datasets like occluded-REID (O-REID), partial-REID (P-REID), and partial-iLIDS (P-iLIDS), clearly showcase the merits of our proposed solution.

Impact Statement—Artificial Intelligence (AI) based surveillance has become an important component of the present day society. One challenging surveillance question is can we automatically find a match of a partially visible person of interest (query image) in a set of partially or fully visible persons (gallery images). Existing methods require heavy computation and/or manual intervention to provide only a partially satisfactory answer. We have achieved state-of-the-art results for the above problem, and that too with less computation. So, our method has much wider applicability, especially, in resource-constrained automated surveillance scenarios.

Index Terms—Knowledge distillation (KD), lightweight framework, occluded and partial re-ID, projective dictionary learning, unary–binary pair.

Manuscript received 26 August 2023; revised 29 October 2023; accepted 15 December 2023. Date of publication 25 December 2023; date of current version 21 June 2024. This article was recommended for publication by Associate Editor C. L. Philip Chen upon evaluation of the reviewers' comments. (Corresponding author: Ananda S. Chowdhury.)

Arindam Sikdar is with Edge Hill University, L39 4QP Ormskirk, U.K. (e-mail: sikdara@edgehill.ac.uk).

Ananda S. Chowdhury is with the Department of Electronics and Telecommunication Engineering, Jadavpur University, Kolkata 700032, India (e-mail: as.chowdhury@jadavpuruniversity).

Digital Object Identifier 10.1109/TAI.2023.3346333

I. INTRODUCTION

PERSON re-identification (re-ID), a surveillance task dealing with matching persons of interest across nonoverlapping views with varying viewpoints, locations, and lightning conditions, has been well studied [1], [2], [3]. Widespread usage of surveillance cameras raised the demand for developing efficient methods for handling more challenging situations like partial and occluded re-ID [4], [5]. Most existing approaches assume full body visibility, which is difficult to guarantee in crowded scenes. Visibility of the person of interest can be adversely affected due to occlusion resulting from surrounding nontarget pedestrian (NTP) and nonpedestrian occlusion (NPO) [2]. Please see Fig. 1 in this context. While NPO partially conceals the visual attributes of the person of interest, NTP alters the visual attributes of the target person. Both these factors are commonly responsible for increasing intraclass and reducing interclass distance. These challenges can be jointly tackled if we can sufficiently rely on the more distinct visual attributes of the target person across different views.

The published works in person re-ID can be broadly categorized into two broad categories, namely, holistic re-ID, and partial and occluded re-ID. Holistic re-ID [6], [7] approaches typically utilize the full body appearance of the person and aim to learn discriminative representations that capture the overall characteristics of the individual. Various techniques, such as hand-crafted features [6], metric learning [4], [5], and deep learning-based methods [8], [9], [10], have demonstrated impressive performance in handling variations in pose, lighting, and appearance for matching individuals across different camera views. However, these methods have limitations when dealing with partial and occluded person images, where only certain parts are visible due to obstructions. Holistic re-ID approaches may not be robust to partial appearances, as they often struggle to effectively capture local part features of limited visible regions with person identity labels available exclusively for the full image. Therefore, specialized techniques are required to address these limitations and improve the accuracy of person re-ID in partial and occluded scenarios. For instance, Zheng et al. [11] proposed a dictionary learning (DL)-based ambiguity-sensitive matching classifier (AMC) that covers different body parts using a sliding window approach to achieve local-to-global matching. Similarly, He et al. [12] developed an alignment-free approach based on deep spatial feature reconstruction (DSR) with sparse coding. Luo et al. [13] incorporated a spatial transformer network (STN) in re-ID frameworks



Fig. 1. Examples illustrating the challenges of occluded re-ID for both (a) NPO and (b) NTPs.

to address the partial body region problem. Sun et al. [14] introduced a visibility-aware part model (VPM) that learns different part regions through self-supervision. Furthermore, Gao et al. [15] developed a pose-guided visible part matching (PVPM) approach to learn pose discriminative features using pose-guided attentions. Despite these advancements, occluded and partial re-ID methods have their limitations. Many approaches in this category heavily rely on outside tools to obtain visual clues of different body parts due to the lack of explicit body part annotations in existing re-ID datasets. Reliance on such external tools easily increases the computational costs of these frameworks. Moreover, most of these methods require manual cropping, which introduce substantial human bias that can restrict the overall re-ID performance. The above strategies may further perform below par when occlusions occur due to other NTPs, leading to part confusion and reduced accuracy.

To address these challenges, we propose a patch-based lightweight DL framework that prioritizes semantically similar parts and reduces reliance on occluded regions when computing the overall score. DL aims to find representative elements (atoms) from a dataset to efficiently represent data through a linear combination of these atoms. This process involves sparse coding and iterative updates, enhancing data representation and compression. Traditional DL methods are computationally expensive due to costly l_0 and l_1 norm computations. However, recent developments in projective DL [16] allow effective representation of discriminative body regions with less computational burden. In contrast to the traditional approach in DL, which involves learning only the synthesis dictionary directly from the signal, projective DL introduces a unique paradigm with two essential components: the analysis dictionary and the synthesis dictionary. The analysis dictionary is designed to efficiently generate discriminative (analytical) codes through linear projection. It is composed of carefully learned basis functions, which serve to transform the original data into a lower-dimensional space. Once this transformation into a lower-dimensional space is achieved, the synthesis dictionary, in conjunction with the analytical code, is applied to compute the sparse representation, resulting in a more precise representation of the original signal. Together, these two dictionaries form a “projective dictionary pair.” In this article, we focus on the task of matching the probe and the gallery samples. This typically involves dealing with data from different views, with the primary shift occurring predominantly along the vertical direction. In our approach, we employ a single projective dictionary

pair, which we term “unary dictionary (UD) pair,” to encode horizontal regions (patches) regardless of the view. Simultaneously, we learn two separate projective dictionary pairs for each of the probe and gallery datasets to encode the vertical regions (patches), considering that occlusions primarily affect this direction, depending on the person’s viewpoint. Given that we are learning two distinct dictionaries for each view separately, we refer to this as “binary dictionary (BD) pairs.” A visual representation of both the unary and binary dictionaries is illustrated in Fig. 2.

We then design an efficient learning mechanism for these dictionaries by leveraging knowledge from high-performing deep frameworks, without explicitly using them during the inference stage. Our approach involves knowledge distillation (KD) [17], where we transfer knowledge from a high complexity deep (teacher) model to a low complexity (lightweight) dictionary (student) model, which is appropriate for fitting a small-sized dataset. To facilitate this learning process, we employ omni-scale network (OSNet) [18] as an auxiliary framework for learning/training dictionaries. OSNet is a recent state-of-the-art deep model capable of synergistically combining features captured at different spatial scales through learned weights. However, we do not require OSNet during inference/testing stage, which results in a lightweight model. Incorporating KD from OSNet leads to a significant performance boost across different re-ID datasets. As a result, our model remains discriminative and avoids the issue of overfitting that other deep models may encounter when dealing with smaller datasets. The major contributions of our work are now summarized below:

- 1) We propose a lightweight unary–binary projective DL (UBDL) model for handling occlusion and partial person re-ID. Specifically, we do the following:
 - a) We apply UD and BD to separately encode the horizontal and vertical components of a person’s appearance from patch features.
 - b) We introduce an adaptive weighting mechanism during BD learning. More weight is given to more prevalent occluded patches in the vertical direction to better address the occlusion challenges.
- 2) We show our unary and binary objective functions to be convex. This is done by introducing additional variables in the original objective functions. Consequently, we analytically derive closed form solutions for both these objective functions. This guarantees that our DL process converges to a global solution, pointing to the optimality of the learned dictionary.
- 3) We develop a knowledge similarity matrix to infuse (deep) OSNet’s insights into the proposed (classical/nondeep) UBDL model. Unlike, a typical Teacher–Student approach in deep models, which employ teacher logits to guide student network training, we propose a similarity matrix with exponentially decreasing weights. This matrix is integrated into DL objectives to curb the impacts of unreliable part region features between probe and gallery samples across two views.

The rest of the article is organized as follows: In Section II, we discuss the related works. Following that, in Section III,

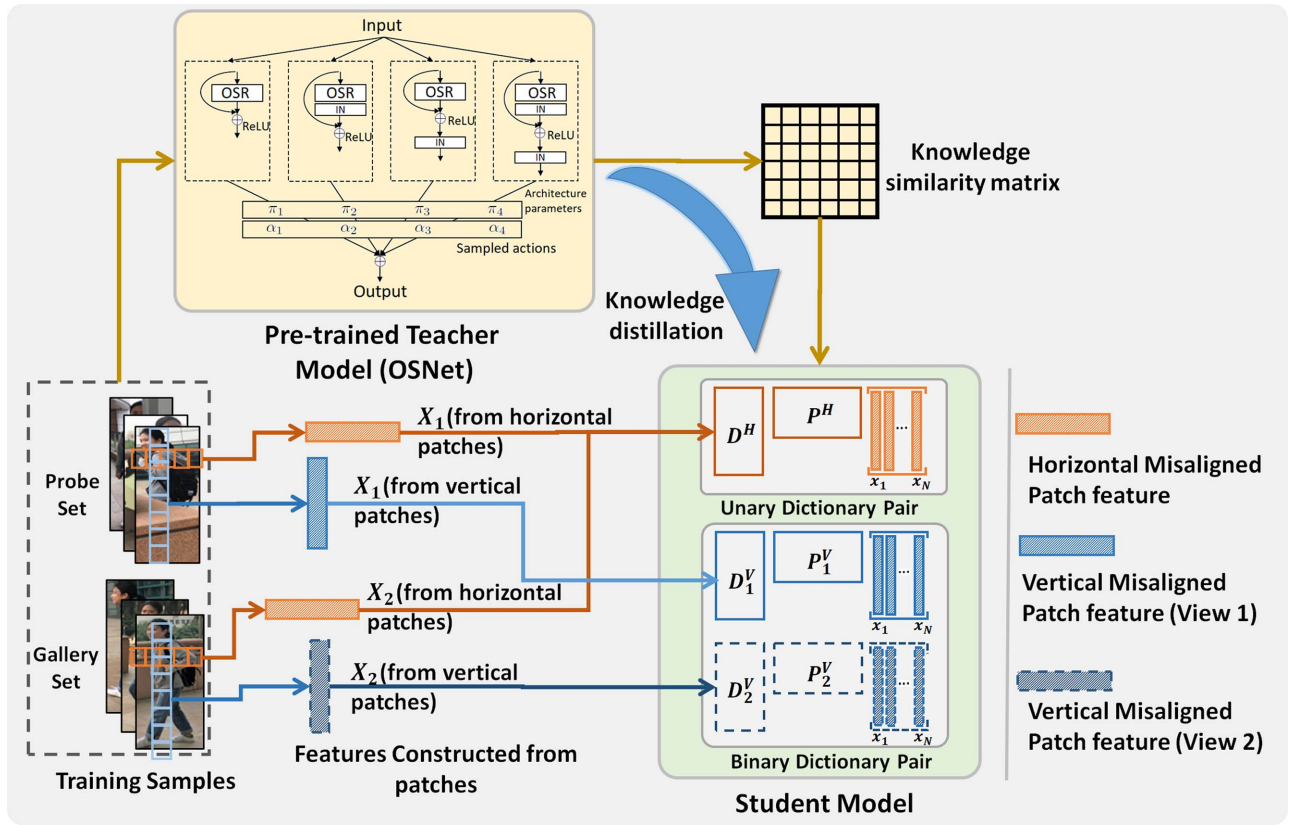


Fig. 2. Illustrative overview of our proposed unary–binary approach learned through KD.

we describe our proposed method. The complexity and convergence analysis are presented in Section IV, while the experimental results are presented in Section V. Finally, in Section VI, we conclude the article and outline directions for future research.

II. RELATED WORKS

In this section, we will provide a comprehensive overview of various methods related to both Holistic re-ID and partial and occluded re-ID. We will also discuss the limitations associated with these approaches.

1) *Holistic Re-ID Approaches*: Methods described in this category are general re-ID approaches that achieved state-of-the-art performance without particularly addressing occlusion problem. These methods have used hand-crafted features [6], [19], metric learning [4], [20], deep learning [8], [21], part-based strategy [22], and patch [23] based approach. For instance, Xu et al. [21] presented a label correction strategy for training with wrongly annotated data. Their novel metric learning mechanism, utilizing cross-entropy and weighted triplet loss, yields more meaningful embeddings for person representations. However, an important limitation of their approach is the potential risk of overfitting when handling noisy data with complex methods. Kalayeh et al. [23] combined multiple part features extracted using human parsing methods. Although semantic segmentation enables precise localization of body parts even under severe pose variations, it may encounter difficulties

in handling partial or heavily occluded individuals. Several attention based methods [8], [24] were utilized to focus on human regions to extract discriminative features. For instance, Zhao et al. [24] utilized elementwise product between person region masks and feature maps to extract part-specific features. This approach improves focus on discriminating regions, increasing the overall matching score for the same person. Nevertheless, the model struggles to generalize well with complex occlusions that obscure important body parts. On the other hand, Li et al. [8] proposed the hard regional attention convolutional neural network (HA-CNN) framework to localize body parts using bounding boxes. While HA-CNN demonstrates advantages in joint learning of attention selection and feature representation, it may not fully address the limitations in handling person re-ID scenarios with arbitrarily aligned person images, large human pose variations, and unconstrained auto-detection errors. While these methods have demonstrated high performance in holistic person matching scenarios, their applicability is restricted in crowded scenes where individuals may be partially or heavily occluded.

2) *Partial and Occluded Re-ID Approaches*: Existing re-ID methods [7] that aim to handle occluded and partial re-ID can be further subdivided into three groups, namely, 1) fixed location based [25], [26]; 2) outside tool-based [27]; and 3) attention based methods [7], [24]. Fixed location based methods typically split the image or output feature maps into several strips at fixed locations before matching. For instance, Sun et al. [26] divided the spatial feature map into six uniform horizontal

strips. The main advantage of these approaches lies in their computational efficiency. However, they disregard the impact of the features from the occluded regions. The second category of methods relies on external tools for human part detection [23] or pose estimation [27] to achieve the re-ID task. The accuracy of such framework depends upon the accurate detection and localization of parts or poses. Also, it requires additional computation costs for the part or pose detection. Furthermore, these methods are most likely to fail in NTP scenario where there is chance of part confusion with other NTPs. Attention based models [7], [28] are quite robust relative to methods in earlier two categories, where more weights are given to body parts belonging to the same person while computing the features. For instance, Gong et al. [7] uses batch attention dropBlock (BA-dropBlock) as local attention system (LAS) without extra supervision to exploit the most salient areas. Nevertheless, due to lack of explicit part supervision, this method may yield sub-optimal performance in scenarios where there is less semantic difference with the surrounding background or when dealing with NTPs. In contrast, quality-aware part models (QPM) [28] are capable of handling challenging occlusion scenarios. Still, the method tends to overlook other nonhuman parts, including personal belongings like backpacks and reticules, which could also provide important clues for identifying target persons. Both the above attention based models achieve better performance but with increasing computational costs.

Our proposed UBDL model with KD addresses the challenge of partial and occluded person re-ID, a feature not present in existing lightweight (mainly nondeep) learning models. We introduce an objective function for learning projective dictionaries with multiple views. Crucially, we incorporate a knowledge similarity matrix into the DL objective, which extends beyond the traditional projective DL model. This addition offers certain advantages, including the incorporation of KD, making our approach more adaptive and capable of learning from diverse sources of information. This mechanism introduces an additional facility to the model, allowing it to transfer knowledge from a (deep) teacher model to a (nondeep) student model. This addition enhances performance, presenting a distinctive feature in our proposed pipeline. Furthermore, through careful consideration of the knowledge similarity matrix, our approach achieves superior results compared to state-of-the-art methods, demonstrating its efficiency and effectiveness.

III. PROPOSED METHOD

In this section, we first state the role of projective DL and utilization of deep KD to improve the DL process. This is followed by the formulation of an objective function for UBDL and a simple decision level score fusion. An overview of the proposed pipeline is shown in Fig. 2. We then show an optimization strategy to properly learn the proposed model.

A. Problem Definition

Let $\mathbf{x}_{ik} \in \mathbb{R}^{d \times 1}$ be the feature vector of dimension d representing k th person from the i th view, where, $k \in \{1, 2, \dots, n\}$ and $i \in \{1, 2\}$. The overall training set for the i th view can then

be represented by the matrix $\mathbf{X}_i = [\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{ik}, \dots, \mathbf{x}_{in}]$, where, $\mathbf{X}_i \in \mathbb{R}^{d \times n}$. Our main goal is to learn a dictionary set that can produce similar encoding for a person undergoing part occlusion between two views. We use the concept of projective dictionary pair [16] to learn more expressive and discriminating sparse representation over conventional DL. In conventional DL, we learn a synthesis Dictionary $\mathbf{D} \in \mathbb{R}^{d \times m}$ that represents a sample set $\mathbf{X}$ via sparse matrix $\mathbf{Z} \in \mathbb{R}^{m \times n}$ such that $\mathbf{X} \approx \mathbf{DZ}$. Such learning involves costly l_1 -norm, whereas a more efficient approach is to find an analysis dictionary $\mathbf{P} \in \mathbb{R}^{m \times d}$ ($m \ll d$) that can generate an analytical code $\mathbf{A} \in \mathbb{R}^{m \times n}$ as $\mathbf{A} = \mathbf{PX}$ such that $\mathbf{X} \approx \mathbf{DA}$. Here, m denotes the dimension of the projected subspace. Thus an analysis ($\mathbf{P}$) and synthesis ($\mathbf{D}$) dictionary can be jointly learned using a framework following [16]

$$\{\mathbf{P}^*, \mathbf{D}^*\} = \underset{\mathbf{P}, \mathbf{D}}{\operatorname{argmin}} \|\mathbf{X} - \mathbf{DPX}\|_F^2 + \Psi(\mathbf{D}, \mathbf{P}, \mathbf{X}) \quad (1)$$

where, $\Psi(\mathbf{D}, \mathbf{P}, \mathbf{X})$ is a discriminative function acting as a misalignment constraint. We further introduce KD to transfer relevant knowledge from a high complexity to a low complexity (lightweight) model via ‘‘Teacher–Student’’ framework [17] for improved learning without overfitting. OSNet [18] acts as a teacher model which is first pretrained on a particular dataset and then used for distilling knowledge into dictionaries (student). OSNet dynamically fuses relevant features from multiple scales through a unified adaptive aggregation mechanism [18]. The deep feature embedding of each person is then used to construct a knowledge similarity matrix $\mathbb{K}$ (as described in the next section). This matrix dynamically assigns higher weights to differences in patch features extracted from similar persons (from different views) after encoding using analysis dictionary $\mathbf{P}$. Dictionaries learned over such weighted feature difference achieve better similarity consensus with OSNet.

B. Formulation of UBDL Objectives

Our main goal is to learn a robust representation that can efficiently identify a person undergoing occlusion between different views. Extracting a single representation may lead to ambiguities caused by background and occluded regions, while relying only on patch space ignores the global structure of a pedestrian in an image [22]. To overcome this challenge, we combined patch features from both horizontal and vertical directions. Specifically, 1) we learned a common dictionary set $\{\mathbf{D}^H, \mathbf{P}^H\}$ in the horizontal direction by combining features from two views (unary DL); and 2) two dictionary sets $\{\mathbf{D}_i^V, \mathbf{P}_i^V\}$ in the vertical direction corresponding to views $i = 1, 2$ (binary DL). Since the height of a person in an upright pose is greater than the width, occlusion has a more significant impact on the vertical direction. To address this challenge, we employ a separate (binary) dictionary in the vertical direction for each pair of views. Accordingly, we suggest proper weights to learn pairwise matching of vertical patch features in the corresponding regions of a person across the two views. This is unlike the horizontal direction, where we learn a common (unary) dictionary across all the views. Features from patches in the horizontal and vertical directions are concatenated to

construct each representation for unary and binary DL respectively. We now propose the UBDL objective function. The objective function for unary DL [having the form of (1)] can be stated as

$$\begin{aligned} \min_{\mathbf{P}, \mathbf{D}} & \|\mathbf{X}_1 - \mathbf{D}^H \mathbf{P}^H \mathbf{X}_1\|_F^2 + \|\mathbf{X}_2 - \mathbf{D}^H \mathbf{P}^H \mathbf{X}_2\|_F^2 \\ & + \lambda_H \|\mathbf{P}^H \mathbf{X}_1 - \mathbf{P}^H \mathbf{X}_2\|_F^2 \\ \text{s.t. } & \|\mathbf{d}_k^H\|_2^2 \leq 1 \end{aligned} \quad (2)$$

where,

$$\mathbb{K} = \begin{bmatrix} e^{-\|\mathbf{f}_{11} - \mathbf{f}_{21}\|_2} & 0 & \dots & 0 \\ 0 & e^{-\|\mathbf{f}_{12} - \mathbf{f}_{22}\|_2} & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & & e^{-\|\mathbf{f}_{1N} - \mathbf{f}_{2N}\|_2} \end{bmatrix}$$

where, $\mathbf{d}^H$ in the above constraint, indicate the columns of the dictionary $\mathbf{D}^H$ normalized to unit norm. The normalization helps to avoid the trivial solution of the above objective when $\mathbf{D}^H \rightarrow \infty$ and $\mathbf{P}^H \mathbf{X} \rightarrow 0$. In the above equation, $\mathbf{f}_{ik}$ is the deep feature embedding of person i from k th view extracted from OSNet (teacher) model. This feature is utilized to construct the knowledge similarity matrix, $\mathbb{K} \in \mathbb{R}^{n \times n}$ indicating a better similarity or closeness between probe and gallery samples. Each diagonal term in the knowledge matrix is a similarity value lying within $(0, 1)$ for a matched pair computed based on the deep feature embedding of OSNet. A true pair will usually yield a higher value closer to 1. This weight will enforce learning of the projection matrix $\mathbf{P}^H$ in such a fashion that the corresponding pair will be encoded more closely in analytical space compared to pairs that are less weighted. Learning the matrix $\mathbf{P}^H$ will in turn influence the learning of synthesis dictionaries $\mathbf{D}^H$ during the alternate optimization step (see Section III-D for further details). It is noteworthy that by fixing the matrix $\mathbb{K}$ to unity ($\mathbf{I}_{n \times n}$) will turn the objective function in (2) into usual DL without KD. In binary DL, we consider all possible combinations of vertical patches within two views while constructing feature set $\mathbf{X}_1$ and $\mathbf{X}_2$. The objective function for Binary DL [having the form of (1)] can be stated as

$$\begin{aligned} \min_{\mathbf{P}, \mathbf{D}} & \|\mathbf{X}_1 - \mathbf{D}_1^V \mathbf{P}_1^V \mathbf{X}_1\|_F^2 + \|\mathbf{X}_2 - \mathbf{D}_2^V \mathbf{P}_2^V \mathbf{X}_2\|_F^2 \\ & + \lambda_V \|\mathbf{P}_1^V \mathbf{X}_1 - \mathbf{P}_2^V \mathbf{X}_2\|_F^2 \\ \text{s.t. } & \|\mathbf{d}_{1k}^V\|_2^2 \leq 1, \|\mathbf{d}_{2k}^V\|_2^2 \leq 1 \end{aligned} \quad (3)$$

where, $\mathbb{W}_{12} = \Omega' \mathbb{K}$, ($\mathbb{W}_{12} \in \mathbb{R}^{n \times n}$) with $\Omega = \text{diag}(\omega_1, \dots, \omega_n)$ and $\Omega' = \text{diag}(\phi(\omega_1), \dots, \phi(\omega_n))$. Here, $\mathbf{d}_{1k}^V$ and $\mathbf{d}_{2k}^V$ are the columns of the dictionary $\mathbf{D}_1^V$ and $\mathbf{D}_2^V$, respectively. The atom of these dictionaries are also normalized to unit norm following similar reasons as (2) and $\phi(z) = 1/(1 + z^2) \in [0, 1]$ is simply a normalization function like sigmoid that instead gives maximum weight (i.e., 1) to smallest distance (i.e., 0) and gradually decreases with increasing distance. Furthermore, $\omega_i = \|\mathbf{P}_1^V \mathbf{x}_{1i} - \mathbf{P}_2^V \mathbf{x}_{2i}\|_2^2, i = 1, \dots, n$ is the importance weight given to relevant vertical patch features with

corresponding views of a person. Also, the weight matrix $\mathbb{W}_{12}$ is adaptive in nature as it depends upon ω_i 's dynamically constructed over patch features in projected subspace. Based on the nature of the objective in (3) the most similar analytical codes of two vertical patches of a person within two views will be weighted close to 1 while with increased dissimilarity it will gradually tend to 0.

C. Fusion and Matching Strategy

In order to perform re-ID matching, we need to compute similarity scores between probe and gallery images. Two separate scores from unary and binary matching are first obtained. These scores are then fused to compute the final score following [22] as described below.

1) *Unary Matching Score*: We begin by computing sparse codes for horizontal features. Specifically, we obtain sparse codes by solving the following equation using the orthogonal matching pursuit (OMP) algorithm [29]

$$\mathbf{z} = \arg \min_{\mathbf{z}} \|\mathbf{x} - \mathbf{D}_{\zeta}^H \mathbf{z}\|^2. \quad (4)$$

In the above equation, $\mathbf{z}$ represents sparse code for the horizontal feature $\mathbf{x}$. A specific support set ζ is extracted from the learned dictionary $\mathbf{D}^H$, and, is denoted by $\mathbf{D}_{\zeta}^H$. This support set is a collection of indices or positions that indicate the dictionary elements (atoms) contributing to the nonzero coefficients in the sparse representation of the signal. For more detailed information on how this sparse representation is computed using OMP algorithm, please refer [29]. Subsequently, the cosine similarity score is calculated between corresponding horizontal components ($h \in \{h_1, h_2, \dots, h_H\}$) of a probe image p and a gallery image g as

$$S_h(p, g) = \frac{\mathbf{z}_p \cdot \mathbf{z}_g}{\|\mathbf{z}_p\| \|\mathbf{z}_g\|}. \quad (5)$$

The overall similarity score between the probe p and gallery g is obtained by summing up all the individual similarity scores of different components: $S_H(p, g) = \sum_h S_h(p, g)$.

2) *Binary Matching Score*: In the binary matching approach, each vertical component feature of the probe image p and gallery image g is sparsely encoded using two separately learned dictionaries, i.e., $\mathbf{D}_1^V$ and $\mathbf{D}_2^V$, employing OMP algorithm, similar to the unary approach. After encoding, cosine similarity scores are computed for all pairwise combinations of the vertical components of the probe ($p \in \{p_1, p_2, \dots, p_k, \dots, p_V\}$) and the gallery ($g \in \{g_1, g_2, \dots, g_V\}$). We can now identify the vertical components between the probe image p and each gallery image g that exhibit higher similarity in their sparse encoding using $\hat{S}_v(p, g) = \arg \min_g S_v(p, g)$, where, S_v assumes the same form as in (4). The overall score between probe p and gallery g is then given by: $S_V(p, g) = \sum_v \hat{S}_v(p, g)$. Finally, we combine the unary and binary matching scores ($S_H(p, g), S_V(p, g)$) over all gallery samples n with a simple fusion strategy in the following way

$$\mathbb{S}(p, g) = S_H(p, g) + \gamma \cdot S_V(p, g) \quad (6)$$

where, $\mathbb{S}$ is overall matching score and γ is tradeoff parameter defined by user. Changes in camera viewpoint cause shifts in body parts within detected bounding boxes, mainly along the vertical direction due to a person's change of orientation about the vertical axis. The proposed similarity score $\hat{S}_v(p, g)$ could still identify visible regions, disregard occluded parts and improve robust matching. Our goal is to determine the best vertical gallery patch with the least distance to each vertical probe patch using $\hat{S}_v(p, g)$. However, such shifts are insignificant along the horizontal direction (S_h computation) since pedestrians typically remain upright. The matching scores are determined based on computed sparse codes using the learned dictionaries $\mathbf{D}^{(\cdot)}$ instead of the analytical code $\mathbf{A}$ (as discussed in Section III-A). This is because the dense analytical representation tends to yield weak matching scores for misaligned patches. Designing a lightweight neural network with KD runs the risk of overfitting, especially, if the amount of data used for distillation is insufficient. Addressing partial or occluded re-id could easily be further challenging. We now show the steps of our solution in Algorithm 1. In our formulation, KD has been effectively integrated into DL objectives.

D. Optimization

We now discuss the optimization strategy of our proposed unary–binary objective functions (2) and (3), respectively.

1) *Solving Objective in (2)*: The objective function of unary DL is nonconvex in nature and is hence hard to solve. We introduce two variables $\mathbf{A}_1^H$ and $\mathbf{A}_2^H$ to relax the objective function in the following manner

$$\begin{aligned} \min_{\substack{\mathbf{P}^H, \mathbf{D}^H, \\ \mathbf{A}_1^H, \mathbf{A}_2^H}} \quad & \|\mathbf{X}_1 - \mathbf{D}^H \mathbf{A}_1^H\|_F^2 + \|\mathbf{X}_2 - \mathbf{D}^H \mathbf{A}_2^H\|_F^2 \\ & + \alpha_H (\|\mathbf{P}^H \mathbf{X}_1 - \mathbf{A}_1^H\|_F^2 + \|\mathbf{P}^H \mathbf{X}_2 - \mathbf{A}_2^H\|_F^2) \\ & + \lambda_H \|(\mathbf{A}_1^H - \mathbf{A}_2^H)\mathbb{K}\|_F^2 \\ \text{s.t. } & \|\mathbf{d}_k^H\|_2^2 \leq 1 \end{aligned} \quad (7)$$

where, α_H is a scalar relaxation constant. Optimization of certain functions involving multiple variables may not be convex when considering all variables simultaneously. However, fixing all variables except the one we intend to optimize can make the function convex (see Section IV). In this case, we are dealing with the minimization of Frobenius norms, where each term involves two variables from the set $(\mathbf{D}^H, \mathbf{P}^H, \mathbf{A}_1^H, \mathbf{A}_2^H)$. As Frobenius norms are quadratic functions, they are indeed convex when each variable is considered in isolation [16].

1) Update steps for $\mathbf{A}_1^H$ and $\mathbf{A}_2^H$: When all other variables are fixed except $\mathbf{A}_1^H$ the objective of (7) reduces to

$$\begin{aligned} \min_{\mathbf{A}_1^H} J_{\mathbf{A}_1^H} = & \|\mathbf{X}_1 - \mathbf{D}^H \mathbf{A}_1^H\|_F^2 + \alpha_H \|\mathbf{P}^H \mathbf{X}_1 - \mathbf{A}_1^H\|_F^2 \\ & + \lambda_H \|(\mathbf{A}_1^H - \mathbf{A}_2^H)\mathbb{K}\|_F^2. \end{aligned} \quad (8)$$

Algorithm 1 Unary binary DL for partial and occluded person re-ID

Input: Person feature $\mathbf{X}_i$ from two views $i \in \{1, 2\}$, parameters α, λ, γ

Output: Matching Scores $\mathbb{S}$ (equation 6)

- 1: *Initialize:* $\mathbf{D}^{H(0)}, \mathbf{P}^{H(0)}, \mathbf{D}_i^{V(0)}, \mathbf{P}_i^{V(0)}$ as random matrices with unit Frobenius norm; $\alpha \leftarrow 0.05, \lambda \leftarrow 0.005, \gamma \leftarrow 0.6$
 - 2: **while** not converge **do**
 - 3: $t \leftarrow t + 1$
 /* Updating unary dictionary components */
 - 4: Update $\mathbf{A}_1^{H(t)}$ and $\mathbf{A}_2^{H(t)}$ using 9 and 10 respectively
 - 5: Update $\mathbf{P}^{H(t)}$ using 12
 - 6: Update $\mathbf{D}^{H(t)}$ using 13
 /* Updating binary dictionary components */
 - 7: Update $\mathbf{A}_1^{V(t)}$ and $\mathbf{A}_2^{V(t)}$ using 15 and 16 respectively
 - 8: Update $\mathbf{P}_1^{V(t)}$ and $\mathbf{P}_2^{V(t)}$ using 18 and 19 respectively
 - 9: Update $\mathbf{D}_i^{V(t)}$ similar to step 6
 - 10: **end while**
 - 11: Extract probe and gallery features from test set (see section III-C)
 - 12: Encode features using learned dictionaries $\mathbf{D}^{H*}$ and $\mathbf{D}_i^{V*}$
 - 13: Compute cosine similarity between probe and gallery sets
 - 14: Fuse and obtain matched score $\mathbb{S}$ (using 6)
-

By setting $\partial J_{\mathbf{A}_1^H} / \partial \mathbf{A}_1^H = 0$, the closed form solution for $\mathbf{A}_1^H$ is obtained as

$$\mathbf{A}_1^{H*} = \left(\mathbf{D}^{H^T} \mathbf{D}^H + \alpha_H \mathbf{I} + \lambda_H \mathbf{B}_1^H \right)^{-1} \left(\mathbf{D}^{H^T} \mathbf{X}_1 + \lambda_H \mathbf{A}_2^H \mathbb{K}^2 + \alpha_H \mathbf{P}^H \mathbf{X}_1 \right). \quad (9)$$

Following similar steps, a closed form solution for $\mathbf{A}_2^H$ can also be obtained as

$$\mathbf{A}_2^{H*} = \left(\mathbf{D}^{H^T} \mathbf{D}^H + \alpha_H \mathbf{I} + \lambda_H \mathbf{B}_2^H \right)^{-1} \left(\mathbf{D}^{H^T} \mathbf{X}_2 + \lambda_H \mathbf{A}_1^H \mathbb{K}^2 + \alpha_H \mathbf{P}^H \mathbf{X}_2 \right) \quad (10)$$

where, $\mathbf{B}_i^H = \mathbf{A}_i^H \mathbb{K}^2 \mathbf{A}_i^{H+}$ and $\mathbf{A}_i^{H+}$ is the Moore–Penrose inverse of nonsquare matrix $\mathbf{A}_i^H$. Please see Appendix A for derivation of solutions of (9) and (10).

2) Update steps of $\mathbf{P}^H$: When all other variables fixed except $\mathbf{P}^H$ the objective of (7) reduces to

$$\begin{aligned} \min_{\mathbf{P}^H} J_{\mathbf{P}^H} = & \|\mathbf{P}^H \mathbf{X}_1 - \mathbf{A}_1^H\|_F^2 + \|\mathbf{P}^H \mathbf{X}_2 - \mathbf{A}_2^H\|_F^2 \\ = & \|\mathbf{P}^H \mathbf{X} - \mathbf{A}^H\|_F^2 \end{aligned} \quad (11)$$

where, $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$ and $\mathbf{A}^H = [\mathbf{A}_1^H \ \mathbf{A}_2^H]$ with $\mathbf{X} \in \mathbb{R}^{d \times 2N}$ and $\mathbf{A}^H \in \mathbb{R}^{m \times 2N}$ respectively. By setting $\partial J_{\mathbf{P}^H} / \partial \mathbf{P}^H = 0$, the closed form solution for $\mathbf{P}^H$ is obtained as

$$\mathbf{P}^{H*} = \mathbf{A}^H \mathbf{X}^T (\mathbf{X} \mathbf{X}^T + \epsilon \mathbf{I})^{-1} \quad (12)$$

where, ϵ is a very small number (value set to $10e^{-4}$) to avoid the singularity problem.

3) Update steps of $\mathbf{D}^H$:

$$\begin{aligned} \min_{\mathbf{D}^H} J_{\mathbf{D}^H} &= \|\mathbf{X}_1 - \mathbf{D}^H \mathbf{A}_1^H\|_F^2 + \|\mathbf{X}_2 - \mathbf{D}^H \mathbf{A}_2^H\|_F^2 \\ &= \|\mathbf{X} - \mathbf{D}^H \mathbf{A}^H\|_F^2 \\ \text{s.t. } &\|\mathbf{d}_k^H\|_2^2 \leq 1. \end{aligned} \quad (13)$$

The above objective function is solved using the ADMM approach [16] to obtain $\mathbf{D}^{H*}$.

2) *Solving Objective in (3)*: Similar to objective (2) the objective function of binary DL (3) is reformulated by introducing two variables $\mathbf{A}_1^V$ and $\mathbf{A}_2^V$ as

$$\begin{aligned} \min_{\substack{\mathbf{P}_1^V, \mathbf{D}_1^V, \\ \mathbf{P}_2^V, \mathbf{D}_2^V, \\ \mathbf{A}_1^V, \mathbf{A}_2^V}} &\|\mathbf{X}_1 - \mathbf{D}_1^V \mathbf{A}_1^V\|_F^2 + \|\mathbf{X}_2 - \mathbf{D}_2^V \mathbf{A}_2^V\|_F^2 \\ &+ \alpha_V (\|\mathbf{P}_1^V \mathbf{X}_1 - \mathbf{A}_1^V\|_F^2 + \|\mathbf{P}_2^V \mathbf{X}_2 - \mathbf{A}_2^V\|_F^2) \\ &+ \lambda_V \|(\mathbf{A}_1^V - \mathbf{A}_2^V) \mathbb{W}_{12}\|_F^2 \\ \text{s.t. } &\|\mathbf{d}_{1k}^V\|_2^2 \leq 1, \|\mathbf{d}_{2k}^V\|_2^2 \leq 1. \end{aligned} \quad (14)$$

1) Update steps of $\mathbf{A}_1^V$ and $\mathbf{A}_2^V$

$$\mathbf{A}_1^{V*} = \left(\mathbf{D}_1^{V^T} \mathbf{D}_1^V + \alpha_V \mathbf{I} + \lambda_V \mathbf{B}_1^V \right)^{-1} \left(\mathbf{D}_1^{V^T} \mathbf{X}_1 + \lambda_V \mathbf{A}_2^V \mathbb{W}_{12}^2 + \alpha_V \mathbf{P}_1^V \mathbf{X}_1 \right) \quad (15)$$

$$\mathbf{A}_2^{V*} = \left(\mathbf{D}_2^{V^T} \mathbf{D}_2^V + \alpha_V \mathbf{I} + \lambda_V \mathbf{B}_2^V \right)^{-1} \left(\mathbf{D}_2^{V^T} \mathbf{X}_2 + \lambda_V \mathbf{A}_1^V \mathbb{W}_{12}^2 + \alpha_V \mathbf{P}_2^V \mathbf{X}_2 \right) \quad (16)$$

where, $\mathbf{B}_i^V = \mathbf{A}_i^V \mathbb{W}_{12}^2 \mathbf{A}_i^{V+}$ and $\mathbf{A}_i^{V+}$ is the Moore–Penrose inverse of nonsquare matrix $\mathbf{A}_i^V$. The solutions of (15) and (16) can be derived following a similar approach as described in Appendix A by just replacing the matrix $\mathbb{K}$ with $\mathbb{W}_{12}$.

2) Update steps of $\mathbf{P}_1^V$ and $\mathbf{P}_2^V$: Due to the introduction of $\mathbb{W}_{12}$ the reduced objective function for $\mathbf{P}_1^V$ is different from (11) as

$$\min_{\mathbf{P}_1^V} J_{\mathbf{P}_1^V} = \alpha_V \|\mathbf{P}_1^V \mathbf{X}_1 - \mathbf{A}_1^V\|_F^2 + \lambda_V \|(\mathbf{A}_1^V - \mathbf{A}_2^V) \mathbb{W}_{12}\|_F^2. \quad (17)$$

Note the second term appears in the above equation $\mathbb{W}$ is dependent on $\mathbf{P}_1^V$. By setting $\partial J_{\mathbf{P}_1^V} / \partial \mathbf{P}_1^V = 0$, the closed form solution for $\mathbf{P}_1^V$ is obtained as

$$\mathbf{P}_1^{V*} = (\alpha_V \mathbf{A}_1^V \mathbf{X}_1^T + \lambda_V \mathbf{P}_2^V \mathbf{M}_{21}) (\alpha_V \mathbf{X}_1 \mathbf{X}_1^T + \lambda_V \mathbf{M}_{11})^{-1}. \quad (18)$$

Similarly, the closed form solution for $\mathbf{P}_2^V$ is:

$$\mathbf{P}_2^{V*} = (\alpha_V \mathbf{A}_2^V \mathbf{X}_2^T + \lambda_V \mathbf{P}_1^V \mathbf{M}_{12}) (\alpha_V \mathbf{X}_2 \mathbf{X}_2^T + \lambda_V \mathbf{M}_{22})^{-1} \quad (19)$$

where, $\mathbf{M}_{12} = \sum_k \|\mathbf{a}_{1k} - \mathbf{a}_{2k}\|_2^2 \phi'(\|\mathbf{P}_1^V \mathbf{x}_{1k} - \mathbf{P}_2^V \mathbf{x}_{2k}\|_2^2) \mathbf{x}_{1k} \mathbf{x}_{2k}^T$ and ϕ' is the derivative of ϕ described in Section III-B.

3) Update steps of $\mathbf{D}_1^V$ and $\mathbf{D}_2^V$: The objective function for updating $\mathbf{D}_1^V$ and $\mathbf{D}_2^V$ is similar to objective (13) with the only difference that $\mathbf{D}_1^V$ and $\mathbf{D}_2^V$ are independently updated.

Our objective functions are meticulously crafted to enable the derivation of closed form solutions for updating both unary

[see (7)] and BD [see (14)] components within our optimization framework. By selectively fixing the values of certain variables, we are able to accurately estimate distinct UD components, such as $\mathbf{P}^H$, $\mathbf{A}_1^H$, and $\mathbf{A}_2^H$. These updates are realized through a systematic application of matrix calculus, as elaborated in Appendix A (for a more detailed breakdown of the mathematical derivations). While dealing with the BD components for both the probe and gallery dictionaries, represented by $\{\mathbf{P}_1^V, \mathbf{D}_1^V\}$ and $\{\mathbf{P}_2^V, \mathbf{D}_2^V\}$, respectively (as described in Appendix B), we employ a similar methodology. Even though we maintain separate sets of dictionary pairs, we are able to derive analytical solutions by iteratively updating each variable while keeping the others fixed. These analytical solutions are expressed as mathematical equations, granting us an in-depth understanding of the dictionary update process. The deterministic nature of the closed form solutions ensures proper optimization of the proposed objective functions. This is especially critical in DL, where the quality of the dictionary directly influences data representation. Unlike many iterative methods that face challenges related to convergence, such as converging to local minima or experiencing slow convergence rates, closed form solutions consistently provide reliable results without requiring any complex initialization. Notably, in our optimization problem, all minimization objectives have a lower bound of 0 due to the quadratic nature of the function. So, our DL process is guaranteed to converge to a global solution [16], pointing to the optimality of the learned dictionary. On the contrary, objective functions that solely rely on iterative methods might only converge to local optima, without the capacity to ensure global optimality. The closed form solutions, therefore, empower us with precise and reliable optimization results while ensuring global convergence in our DL process.

IV. COMPLEXITY AND CONVERGENCE ANALYSIS

In this section, we provide the time complexity and convergence analysis of our proposed DL algorithm.

A. Complexity

In training phase of UBDL, we are simultaneously optimizing unary and binary objectives as given by (7) and (14), respectively. To solve for unary objective in (7), we require computing the closed form solution for $\mathbf{A}_i^H$ (i th view), $\mathbf{P}^H$ and $\mathbf{D}^H$ alternatively in an iterative fashion (see Algorithm 1). The complexities of updating $\mathbf{A}_i^H$, $\mathbf{P}^H$, and $\mathbf{D}^H$ are $\mathcal{O}(m^3 + m^2d + mdn)$, $\mathcal{O}(d^3 + md^2 + mdn)$, and $\mathcal{O}(W(m^3 + m^2d + d^2m + mdn))$, respectively, where, W is the total iteration for updating $\mathbf{D}^H$ in ADMM algorithm [see (13)]. These complexities are found to be same as [16]. The time complexity for updating $\mathbf{P}_i^V$ and $\mathbf{D}_i^V$ in binary objective (14) is also found to be the same as $\mathbf{P}^H$ and $\mathbf{D}^H$, respectively. However, the time complexity for updating $\mathbf{A}_i^V$ is found to be $\mathcal{O}(m^3 + m^2d + m^2n^2 + mdn)$ different from that of $\mathbf{A}_i^H$. Here, we have an additional term of m^2n^2 which comes from the computation of $\mathbf{B}_i$ in (15) and (16). In general, the feature dimension (d) of horizontal and vertical patches is much larger than the number of training samples (n)

and dictionary atoms (m) due to feature concatenation (see Section III-B). Here, the major computational burden is on updating $\mathbf{P}^H$ and $\mathbf{P}_i^V$ which involves the inversion of $d \times d$ matrix $(\mathbf{X}\mathbf{X}^T + \epsilon\mathbf{I})$ and $(\alpha_V \mathbf{X}_i \mathbf{X}_i^T + \lambda_V \mathbf{M}_{ii})$, respectively. So, our proposed DL (training) solution has an overall complexity of $O(d^3)$, which is significantly lower than the $O(d^5)$ complexity of a typical feed-forward neural network (composed of d layers, each with d neurons) during training (back-propagation) [30]. In contrast, during the inference stage, our model's computational complexity is derived from the sparse representation using OMP, which has a complexity of $O(d^3)$ (see Section III-C). Notably, this complexity is similar to that of a feed-forward neural network during the forward pass. Therefore, even when training and inference are considered together, the overall complexity of our model is lower than that of a typical feed-forward neural network. This observation further establishes the lightweight (low complexity) nature of our model.

B. Convergence

In general, a function $f(x)$ is convex if it satisfies the triangle inequality for any two points x_1 and x_2 as $f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2)$ where $t \in [0, 1]$. Based on [31] our objective function in (7) (say J_1) can be shown as tri-convex for $\{(\mathbf{D}^H, \mathbf{P}^H), \mathbf{A}_1^H, \mathbf{A}_2^H\}$ in the following manner. By fixing $\mathbf{A}_1^H$ and $\mathbf{A}_2^H$ the variables $\mathbf{D}^H$ and $\mathbf{P}^H$ are separable as $J_1(\mathbf{D}^H) = (\|\mathbf{X}_1 - \mathbf{D}^H \mathbf{A}_1^H\|_F^2 + \|\mathbf{X}_2 - \mathbf{D}^H \mathbf{A}_2^H\|_F^2)$ and $J_1(\mathbf{P}^H) = (\|\mathbf{P}^H \mathbf{X}_1 - \mathbf{A}_1^H\|_F^2 + \|\mathbf{P}^H \mathbf{X}_2 - \mathbf{A}_2^H\|_F^2)$ and therefore can be jointly optimized. Also each of these functions separately satisfies the convex property which also makes the summation a convex function. Here, the variables $\mathbf{A}_1^H$ and $\mathbf{A}_2^H$ can separately form two convex objective functions if all other variables are kept fixed and are thus optimized independently. This makes the overall objective function in (7) tri-convex. By similar reasoning, the objective function in (14) can be shown quad-convex for the set $\{(\mathbf{D}_1^V, \mathbf{P}_1^V), (\mathbf{D}_2^V, \mathbf{P}_2^V), \mathbf{A}_1^V, \mathbf{A}_2^V\}$. By first fixing $(\mathbf{D}_2^V, \mathbf{P}_2^V)$, $\mathbf{A}_1^V$ and $\mathbf{A}_2^V$, we can jointly form a convex objective function for variables $\mathbf{D}_1^V$ and $\mathbf{P}_1^V$ as $\|\mathbf{X}_1 - \mathbf{D}_1^V \mathbf{A}_1^V\|_F^2$ and $\mathbf{J}_{P_1^V}$ [see (17)] are both convex in nature. By similar argument, $\mathbf{D}_1^V$ and $\mathbf{P}_1^V$ can also be jointly optimized, whereas, $\mathbf{A}_1^V$ and $\mathbf{A}_2^V$ separately form a convex function. Each of these objective functions are continuously differentiable and monotonically increasing with a lower bound of 0 and are therefore guaranteed to converge to a global minimum.

V. EXPERIMENTS

In this section, we first present a brief description of the various datasets used, evaluation measures and implementation details. This is followed by a detailed comparative analysis with various state-of-the-art approaches and ablation studies.

A. Datasets and Evaluation Metrics

We have experimented with three benchmark re-ID datasets having various partial and occluded visibility of pedestrians, namely, occluded-REID (O-REID) [32], partial-REID

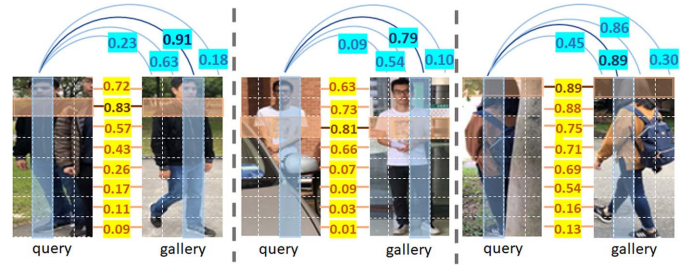


Fig. 3. Similarity scores for patch features in horizontal (in orange) and vertical (in blue) direction for some NPO and NTP cases. Similarities between corresponding and pairwise locations are computed for horizontal and vertical direction, respectively. The highlighted regions and scores are best matches.

(P-REID) [11], and partial-iLIDS (P-iLIDS) [12]. O-REID [32] is a dataset on occluded re-ID with 2000 images belonging to 200 distinct persons. The dataset is composed of 5 full body and 5 occluded samples for each persons. P-REID [11] dataset samples were collected at university campus consisting of 600 images from 60 pedestrians. Each person has 5 occluded and 5 full body images with viewpoint and background variations. P-iLIDS [12] dataset was constructed from airport scene containing 238 images from 119 identities with frequent occlusion occurring from individuals and luggages. Each person contributes to 1 full body and 1 occluded image to the dataset. For both P-REID and P-iLIDS datasets, all probe samples are occluded while all gallery samples are holistic images. All the images are normalized to standard resolution following [22] before processing. We have followed the standard measures for performance evaluation, *i.e.*, 1) mean average precision (mAP) and 2) Rank-k recognition rates, and reported the same in the article.

B. Implementation Details

The representative feature vector of each person is formed by combining features from dense grid patches [22]. To achieve this, each image is divided into patches of size 10×10 with a grid step of 5. For each patch, a 32-bin color histogram and a 128-dimensional scale-invariant feature transform (SIFT) feature are computed for each LAB channel. Additionally, color histograms are computed from down-sampled versions of patches (by factors 0.75 and 0.5) to enhance color information robustness. The concatenation of color histograms and SIFT features results in a patch dimension of $d = 32 \times 3 \times 3 + 128 \times 3 = 672$, which is then normalized to unit length. Subsequently, for constructing vertical and horizontal features, these patch features are further concatenated in their respective directions (see Fig. 3). The deep feature embedding (512-dimensional) is extracted from the OSNet's fully connected (fc) layer as described in Section III-B. Notably, we have only used pretrained (on ImageNet) OSNet (available at https://kaiyangzhou.github.io/deep-person-reid/MODEL_ZOO) and fine-tuned on specific partial/occluded re-ID datasets for KD following [18]. No additional/other re-ID datasets were used for pretraining OSNet to ensure fairness in DL. During evaluation, we solely used

TABLE I
COMPARISONS OF MATCHING RATES (%)
ON OCCLUDED DATASETS

Method	O-REID	
	R@1	mAP
DSR [12]	72.8	62.8
FPR [33]	78.3	68.0
PVPM [15]	70.4	61.2
HOReID [34]	80.3	70.2
PAT [1]	81.6	72.1
ViT Baseline [35]	81.2	76.7
TransReID [35]	70.2	67.3
FED [2]	86.3	79.3
BPBReID [36]	82.9	75.2
RGANet [37]	86.4	80.0
UBDL	87.5	78.2
UBDL + KD	89.3	80.6

Note: Bold entries indicate best Rank-1 and mAP values.

TABLE II
COMPARISONS OF MATCHING RATES (%) ON PARTIAL
REID DATASETS

Method	P-REID		P-iLIDS	
	R@1	R@3	R@1	R@3
AMC + SWM [11]	21.0	32.8	37.3	46.0
DSR [12]	58.8	67.2	50.7	70.0
DCR [38]	60.5	69.7	52.0	67.5
STNReID [13]	54.6	71.3	66.7	80.3
VPM [14]	65.5	74.8	67.7	81.9
HOReID [34]	72.6	86.4	85.3	91.0
QPM [28]	77.3	85.7	81.7	88.0
RGANet [37]	87.2	93.5	77.0	87.6
UBDL	78.02	87.3	84.4	90.5
UBDL + KD	79.3	89.4	85.7	91.3

Note: Bold entries indicate best Rank-1 and Rank-3 values.

the learned dictionary for computing person matching scores. OSNet only acted as an auxiliary teacher model that was used to guide the DL process through KD. In the UBDL objective function (2) and (3), the parameters used are α_H , α_V , λ_H , and λ_V . α_H and α_V result from the convex formulation of projective dictionary pair objectives and are set to 0.05 based on [16]. Additionally, λ_H and λ_V are set to 0.005 according to [22]. Moreover, there's a user parameter γ (6) balancing the tradeoff between unary and binary matching scores, set to 0.6 based on experimental findings.

C. Performance Comparisons

In this section, we evaluate our method against several state-of-the-art approaches in partial and occluded re-ID, as presented in Tables I and II. We have now incorporated two relevant works from 2023 in our article [36], [37].

1) *Results on Occluded Re-ID Dataset:* Our evaluation results for O-ReID, as presented in Table I, exhibits exceptional performance when compared to several recent methods, including DSR [12], feature pyramid reconstruction (FPR) [33], PVPM [15], high-order re-identification (HOReID) [34], PAT [1], ViT Baseline [35], TransReID [35], FED [2], body part based re-identification (BPBReID) [36], and region generation and assessment (RGANet) [37]. When KD is incorporated in the

DL process, the performance improvements become significant. We attain the highest Rank-1 accuracy of 89.3% and the highest mAP of 80.6%, outperforming the closest competitor, RGANet, by 2.9% in Rank-1 and by 0.6% in mAP. Even without KD, our approach demonstrates competitive performance, with a slight 1.8% lag behind RGANet in mAP. Notably, our method significantly outperforms TransReID by 19.1% in Rank-1 and 13.3% in mAP, despite TransReID being an exclusive transformer-based framework.

2) *Results on Partial Re-ID Datasets:* Table II shows the performance comparison of our approach with AMC + sliding window matching (SWM) [11], DSR [12], deep spatial pyramid feature collaborative reconstruction (DCR) [38], spatial transformer network based re-identification (STNReID) [13], VPM [14], HOReID [34], QPM [28], and RGANet [37] on P-REID and P-iLIDS datasets. On the P-REID dataset, our approach excels, achieving an impressive Rank-1 score of 79.3% when employing KD, placing it among the top performers in this category. Additionally, we present a remarkable Rank-3 result of 89.4%, surpassing several other methods. In the P-iLIDS dataset, our approach maintains its supremacy, reaching the highest Rank-1 score of an impressive 85.7%. While the RGANet secures the highest Rank-1 and Rank-3 values of 87.2% and 93.5% on the P-REID dataset, our method is a close second. On the P-iLIDS dataset, the RGANet method achieves a commendable Rank-1 of 77.0% and mAP of 87.6%. However, it is crucial to acknowledge that our approach secures the highest Rank-1 score on this dataset, an exceptional 85.7%, underscoring its robustness and effectiveness. Furthermore, our method excels by attaining the highest Rank-3 score of 91.3% on the P-iLIDS dataset, clearly demonstrating its superior ability to navigate complex scenarios in partial person re-ID.

In contrast to existing methods, our lightweight yet highly effective approach for occluded and partial person re-ID clearly stands out. While DSR and FPR rely on complex alignment using convolutional networks and reconstruction to handle occlusions, our method simplifies the process. Unlike TransReID, which focuses on colocated positions, neglecting occluded parts, our method maintains robustness without the complexity of transformers, prioritizing unoccluded parts even in misaligned positions. Moreover, our approach addresses the challenges associated with NPOs and NTPs), explicitly identifying and leveraging unoccluded misaligned parts, thus enhancing matching scores. Unlike BPBReID, which relies on explicit body part annotations, our method is versatile. It does not require specific body part annotations and relies on KD instead. Comparing our approach to others reveals distinctive advantages. AMC + SWM and DSR rely on a combination strategy, while we integrate horizontal and vertical patch feature information, enhancing global context. DCR, using information from different spatial resolutions, may introduce noise. In contrast, our method learns a granular patch-based dictionary for effective matching of unoccluded regions. QPM assigns quality scores to body parts, which depends on annotation accuracy and availability of part information. Our method does not have such constraints. In conclusion, our lightweight yet powerful solution

TABLE III
RANK-1 RECOGNITION RATE (%) ON DIFFERENT COMPONENTS
OF DICTIONARY LEARNING OVER VARIOUS DATASETS

Methods	O-REID	P-REID	P-iLIDS
Unary	72.76	69.5	80.33
Binary	78.52	76.6	82.67
Unary-Binary	87.5	78.02	84.4
Unary + KD	83.2	75.3	82.1
Binary + KD	86.4	77.2	83.2
Unary-Binary + KD	89.3	79.3	85.7

Note: Bold entries indicate best Rank-1 values.

excels in handling occlusions, making it ideal for real-world re-ID scenarios. Our results on specific datasets show modest improvements over existing state-of-the-art techniques. Notably, our dictionaries typically contain fewer than 10^5 parameters, which is significantly fewer than ResNet-18 [4], with around 11.2 million parameters.

D. Ablation Study

We have conducted an ablation study with different components of dictionary, i.e., unary and binary, and that too with and without KD (see Table III). First, a clear trend of improvement can be observed with the fused similarity scores of unary and BD components over the individual dictionary components. We achieve decent Rank-1 scores of 87.5%, 78.02%, and 84.4% without KD and an improved performance of 89.3%, 79.3%, and 85.7% with KD on O-REID, P-REID, and P-iLIDS, respectively. It can be observed that binary component has consistently performed better compared to unary in all cases, which clearly indicates the more discriminative feature construction from vertical patches. Also it is evident from results that improvement due to KD is more in case of larger O-REID datasets compared to P-REID and P-iLIDS dataset due to better generalization of pretrained teacher framework.

VI. CONCLUSION

In this article, we presented a lightweight learning framework for handling occluded and partial re-ID challenges. The framework involved projective DL and KD from a deep learning model. By learning unary and BD pairs, we encoded horizontal patches and prioritized relevant vertical patches for improved occlusion handling. The final re-ID matching score was obtained by combining unary and binary matching scores using an adaptive weighting mechanism. Extensive experiments on three benchmark datasets demonstrated the efficacy of our approach.

Our approach works very well when coupled with the deep OSNet model, but it is essential to mention that this knowledge transfer may not suit all deep learning architectures universally. Its performance might dip when applied to some alternative deep architectures. Furthermore, our BD within the UBDL framework excels at capturing correspondence across different viewpoints. However, it can yield below par results in scenarios with extreme viewpoint variations on nonstandard camera angles. As part of the future work, we

will address these challenges to extend our solution to more different scenarios.

APPENDIX A

PROOF OF $\mathbf{A}_1^H$ (9) AND $\mathbf{A}_2^H$ (10)

For updating $\mathbf{A}_1^H$, we need to find the derivative of objective (8) with respect to $\mathbf{A}_1^H$. For better readability, first let us ignore the superscript H in objective (8) and simply represent $\mathbf{A}_1^H$ by $\mathbf{A}_1$. Now, by ignoring the term which is not depending on $\mathbf{A}_1$ the objective (8) can be re-written as

$$J_{\mathbf{A}_1} = \|\mathbf{X}_1 - \mathbf{D}_1 \mathbf{A}_1\|_F^2 + \alpha \|\mathbf{P}_1 \mathbf{X}_1 - \mathbf{A}_1\|_F^2 + \lambda \|(\mathbf{A}_1 - \mathbf{A}_2) \mathbb{K}\|_F^2 \quad (20)$$

Now, we can find $\partial J_{\mathbf{A}_1} / \partial \mathbf{A}_1$ using matrix calculus as

$$\frac{\partial J_{\mathbf{A}_1}}{\partial \mathbf{A}_1} = -2\mathbf{D}_1^T (\mathbf{X}_1 - \mathbf{D}_1 \mathbf{A}_1) - 2\alpha (\mathbf{P}_1 \mathbf{X}_1 - \mathbf{A}_1) + 2\lambda (\mathbf{A}_1 \mathbb{K} - \mathbf{A}_2 \mathbb{K}) \mathbb{K}^T. \quad (21)$$

By setting $\partial J_{\mathbf{A}_1} / \partial \mathbf{A}_1 = 0$ and expanding all terms, we have

$$\begin{aligned} &\Rightarrow \mathbf{D}_1^T \mathbf{X}_1 - \mathbf{D}_1^T \mathbf{D}_1 \mathbf{A}_1 + \alpha \mathbf{P}_1 \mathbf{X}_1 \\ &\quad - \alpha \mathbf{A}_1 - \lambda \mathbf{A}_1 \mathbb{K}^2 + \lambda \mathbf{A}_2 \mathbb{K}^2 = 0 \\ &\Rightarrow \mathbf{D}_1^T \mathbf{D}_1 \mathbf{A}_1 + \alpha \mathbf{A}_1 + \lambda \mathbf{A}_1 \mathbb{K}^2 \\ &\quad = (\mathbf{D}_1^T \mathbf{X}_1 + \alpha \mathbf{P}_1 \mathbf{X}_1 + \lambda \mathbf{A}_2 \mathbb{K}^2). \end{aligned} \quad (22)$$

Please note that here $\mathbb{K}^T = \mathbb{K}$ is a diagonal matrix. Now, to take $\mathbf{A}_1$ as a common from LHS of the above equation we need to compute $\mathbf{B}_1$ such that $\mathbf{B}_1 \mathbf{A}_1 \approx \mathbf{A}_1 \mathbb{K}^2$. Thus, $\mathbf{B}_1 = \mathbf{A}_1 \mathbb{K}^2 \mathbf{A}_1^+$, where, $\mathbf{A}_1^+$ is Moore-Penrose inverse of $\mathbf{A}_1$ (since $\mathbf{A}_1$ is a nonsquare matrix). Thus (22) can be re-written as

$$\begin{aligned} &\Rightarrow \mathbf{D}_1^T \mathbf{D}_1 \mathbf{A}_1 + \alpha \mathbf{A}_1 + \lambda \mathbf{B}_1 \mathbf{A}_1 \\ &\quad = (\mathbf{D}_1^T \mathbf{X}_1 + \alpha \mathbf{P}_1 \mathbf{X}_1 + \lambda \mathbf{A}_2 \mathbb{K}^2) \\ &\Rightarrow \mathbf{A}_1^* = (\mathbf{D}_1^T \mathbf{D}_1 + \alpha \mathbf{I} + \lambda \mathbf{B}_1)^{-1} \\ &\quad (\mathbf{D}_1^T \mathbf{X}_1 + \alpha \mathbf{P}_1 \mathbf{X}_1 + \lambda \mathbf{A}_2 \mathbb{K}^2). \end{aligned} \quad (23)$$

In a similar fashion, we can compute $\mathbf{A}_2^*$ as

$$\mathbf{A}_2^* = (\mathbf{D}_2^T \mathbf{D}_2 + \alpha \mathbf{I} + \lambda \mathbf{B}_2)^{-1} (\mathbf{D}_2^T \mathbf{X}_2 + \alpha \mathbf{P}_2 \mathbf{X}_2 + \lambda \mathbf{A}_1 \mathbb{K}^2) \quad (24)$$

where, $\mathbf{B}_2 = \mathbf{A}_2 \mathbb{K}^2 \mathbf{A}_2^+$. Please note that both closed form solutions of $\mathbf{A}_1^*$ (23) and $\mathbf{A}_2^*$ (24) are dependent on matrices $\mathbf{A}_1$ and $\mathbf{A}_2$. We have used updated $\mathbf{A}_1$ and $\mathbf{A}_2$ from previous iterations.

APPENDIX B

PROOF OF $\mathbf{P}_1^V$ (18) AND $\mathbf{P}_2^V$ (19)

For updating $\mathbf{P}_1$, we need to find the derivative of objective (17) with respect to $\mathbf{P}_1$. For better readability, we are ignoring the superscript term V in objective (17) and represent $\mathbf{P}_1^V$ and $\mathbf{P}_2^V$ by $\mathbf{P}_1$ and $\mathbf{P}_2$ respectively. Thus we can compute $\partial J_{\mathbf{P}_1} / \partial \mathbf{P}_1$ as

$$\frac{\partial J_{\mathbf{P}_1}}{\partial \mathbf{P}_1} = 2\alpha (\mathbf{P}_1 \mathbf{X}_1 - \mathbf{A}_1) \mathbf{X}_1^T + \lambda \frac{\partial}{\partial \mathbf{P}_1} \|(\mathbf{A}_1 - \mathbf{A}_2) \mathbb{W}_{12}\|_F^2. \quad (25)$$

To solve the above equation first let us solve for 2nd term $\partial Q / \partial \mathbf{P}_1$ where $Q = \|(\mathbf{A}_1 - \mathbf{A}_2) \mathbb{W}_{12}\|_F^2$. As described in Section III-A, $\mathbf{X}_i = [\mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{ik}, \dots, \mathbf{x}_{in}]$, similarly the analytical codes $\mathbf{A}_i \in \mathbb{R}^{m \times n}$ can also be represented as $\mathbf{A}_i = [\mathbf{a}_{i1}, \mathbf{a}_{i2}, \dots, \mathbf{a}_{ik}, \dots, \mathbf{a}_{in}]$ for i th view with $i \in \{1, 2\}$. Thus we can represent Q in summation form

$$Q = \sum_{k=1}^n \phi(\|\mathbf{P}_1 \mathbf{x}_{1k} - \mathbf{P}_2 \mathbf{x}_{2k}\|_2^2) \cdot \|\mathbf{a}_{1k} - \mathbf{a}_{2k}\|_2^2. \quad (26)$$

Thus, derivation of Q with respect to $\mathbf{P}_1$ can be written as

$$\begin{aligned} \frac{\partial Q}{\partial \mathbf{P}_1} &= \sum_{k=1}^n \|\mathbf{a}_{1k} - \mathbf{a}_{2k}\|_2^2 \cdot \phi'(\|\mathbf{P}_1 \mathbf{x}_{1k} - \mathbf{P}_2 \mathbf{x}_{2k}\|_2^2) \\ &\quad \cdot 2(\mathbf{P}_1 \mathbf{x}_{1k} - \mathbf{P}_2 \mathbf{x}_{2k}) \cdot \mathbf{x}_{1k}^T \\ &= 2 \sum_{k=1}^n g_k \cdot (\mathbf{P}_1 \mathbf{x}_{1k} \mathbf{x}_{1k}^T - \mathbf{P}_2 \mathbf{x}_{2k} \mathbf{x}_{1k}^T) \\ &= 2 \left[\mathbf{P}_1 \sum_{k=1}^n (g_k \cdot \mathbf{x}_{1k} \mathbf{x}_{1k}^T) - \mathbf{P}_2 \sum_{k=1}^n (g_k \cdot \mathbf{x}_{2k} \mathbf{x}_{1k}^T) \right] \\ &= 2 [\mathbf{P}_1 \cdot \mathbf{M}_{11} - \mathbf{P}_2 \cdot \mathbf{M}_{21}] \end{aligned} \quad (27)$$

where, $\phi'(z) = -2z/(1+z^2)^2$ is the derivative of function ϕ as described in Section III-B and $g_k = \|\mathbf{a}_{1k} - \mathbf{a}_{2k}\|_2^2 \cdot \phi'(\|\mathbf{P}_1 \mathbf{x}_{1k} - \mathbf{P}_2 \mathbf{x}_{2k}\|_2^2)$. Here, the term $\mathbf{M}_{ij} = \sum_{k=1}^n g_k \mathbf{x}_{ik} \mathbf{x}_{jk}^T \in \mathbb{R}^{d \times d}$ signifies the summation of outer product over any two samples each belonging to i^{th} and j^{th} view respectively. Now, by substituting (27) in (25) we have

$$\frac{\partial J_{\mathbf{P}_1}}{\partial \mathbf{P}_1} = 2\alpha(\mathbf{P}_1 \mathbf{X}_1 - \mathbf{A}_1) \mathbf{X}_1^T + 2\lambda(\mathbf{P}_1 \mathbf{M}_{11} - \mathbf{P}_2 \mathbf{M}_{21}). \quad (28)$$

Now, by setting $\partial J_{\mathbf{P}_1} / \partial \mathbf{P}_1 = 0$ and expanding terms we have

$$\begin{aligned} &\Rightarrow \alpha \mathbf{P}_1 \mathbf{X}_1 \mathbf{X}_1^T - \alpha \mathbf{A}_1 \mathbf{X}_1^T + \lambda \mathbf{P}_1 \mathbf{M}_{11} - \lambda \mathbf{P}_2 \mathbf{M}_{21} = 0 \\ &\Rightarrow \mathbf{P}_1 (\alpha \mathbf{X}_1 \mathbf{X}_1^T + \lambda \mathbf{M}_{11}) = \alpha \mathbf{A}_1 \mathbf{X}_1^T + \lambda \mathbf{P}_2 \mathbf{M}_{21} \\ &\Rightarrow \mathbf{P}_1^* = (\alpha \mathbf{A}_1 \mathbf{X}_1^T + \lambda \mathbf{P}_2 \mathbf{M}_{21}) (\alpha \mathbf{X}_1 \mathbf{X}_1^T + \lambda \mathbf{M}_{11})^{-1}. \end{aligned} \quad (29)$$

In similar fashion, we can compute $\mathbf{P}_2^*$ by setting $\partial J_{\mathbf{P}_2} / \partial \mathbf{P}_2 = 0$ as

$$\mathbf{P}_2^* = (\alpha \mathbf{A}_2 \mathbf{X}_2^T + \lambda \mathbf{P}_2 \mathbf{M}_{12}) (\alpha \mathbf{X}_2 \mathbf{X}_2^T + \lambda \mathbf{M}_{22})^{-1}. \quad (30)$$

Please note that here the matrix $\mathbf{M}_{ji}$ and $\mathbf{M}_{ij}$ are related by transpose operation as $\mathbf{M}_{ij}^T = \mathbf{M}_{ji}$. This can be easily shown as

$$\begin{aligned} \mathbf{M}_{ij}^T &= \left(\sum_{k=1}^n g_k \mathbf{x}_{ik} \mathbf{x}_{jk}^T \right)^T \\ &= \sum_{k=1}^n g_k (\mathbf{x}_{ik} \mathbf{x}_{jk}^T)^T \\ &= \sum_{k=1}^n g_k \mathbf{x}_{jk} \mathbf{x}_{ik}^T = \mathbf{M}_{ji}. \end{aligned} \quad (31)$$

Thus, in order to compute the solution of (29) and (30) we need to compute only one of $\mathbf{M}_{12}$ and $\mathbf{M}_{21}$.

REFERENCES

- [1] Y. Li, J. He, T. Zhang, X. Liu, Y. Zhang, and F. Wu, "Diverse part discovery: Occluded person re-identification with part-aware transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 2898–2907.
- [2] Z. Wang, F. Zhu, S. Tang, R. Zhao, L. He, and J. Song, "Feature erasing and diffusion network for occluded person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 4754–4763.
- [3] T. Ahmad, L. Jin, X. Zhang, S. Lai, G. Tang, and L. Lin, "Graph convolutional neural network for human action recognition: A comprehensive survey," *IEEE Trans. Artif. Intell.*, vol. 2, no. 2, pp. 128–145, Apr. 2021.
- [4] A. Sikdar and A. S. Chowdhury, "Scale-invariant batch-adaptive residual learning for person re-identification," *Pattern Recognit. Lett.*, vol. 129, pp. 279–286, Jan. 2020.
- [5] A. Sikdar, D. Chatterjee, A. Bhowmik, and A. S. Chowdhury, "Open-set metric learning for person re-identification in the wild," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, 2020, pp. 2356–2360.
- [6] S. Bak and P. Carr, "Person re-identification using deformable patch metric learning," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2016, pp. 1–9.
- [7] X. Gong et al., "Lag-net: Multi-granularity network for person re-identification via local attention system," *IEEE Trans. Multimedia*, vol. 24, pp. 217–229, 2021.
- [8] W. Li, X. Zhu, and S. Gong, "Harmonious attention network for person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 2285–2294.
- [9] Z. Ji, J. Hu, D. Liu, L. Y. Wu, and Y. Zhao, "Asymmetric cross-scale alignment for text-based person search," *IEEE Trans. Multimedia*, vol. 25, pp. 7699–7709, 2023.
- [10] L. Wu et al., "Pseudo-pair based self-similarity learning for unsupervised person re-identification," *IEEE Trans. Image Process.*, vol. 31, pp. 4803–4816, 2022.
- [11] W.-S. Zheng, X. Li, T. Xiang, S. Liao, J. Lai, and S. Gong, "Partial person re-identification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2015, pp. 4678–4686.
- [12] L. He, J. Liang, H. Li, and Z. Sun, "Deep spatial feature reconstruction for partial person re-identification: Alignment-free approach," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 7073–7082.
- [13] H. Luo, W. Jiang, X. Fan, and C. Zhang, "STNReID: Deep convolutional networks with pairwise spatial transformer networks for partial person re-identification," *IEEE Trans. Multimedia*, vol. 22, no. 11, pp. 2905–2913, Nov. 2020.
- [14] Y. Sun et al., "Perceive where to focus: Learning visibility-aware part-level features for partial person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 393–402.
- [15] S. Gao, J. Wang, H. Lu, and Z. Liu, "Pose-guided visible part matching for occluded person reid," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11 744–11 752.
- [16] S. Gu, L. Zhang, W. Zuo, and X. Feng, "Projective dictionary pair learning for pattern classification," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2014, pp. 793–801.
- [17] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, pp. 1789–1819, Jun. 2021.
- [18] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, "Learning generalisable omni-scale representations for person re-identification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, p. 5056, Sep. 2022.
- [19] S. Liao, Y. Hu, X. Zhu, and S. Z. Li, "Person re-identification by local maximal occurrence representation and metric learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 2197–2206.
- [20] U. K. Dutta, M. Harandi, and C. C. Sekhar, "Unsupervised deep metric learning via orthogonality based probabilistic loss," *IEEE Trans. Artif. Intell.*, vol. 1, no. 1, pp. 74–84, Aug. 2020.
- [21] F. Xu, B. Ma, H. Chang, and S. Shan, "PRDP: Person reidentification with dirty and poor data," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11 014–11 026, Oct. 2022.
- [22] S. Li, M. Shao, and Y. Fu, "Person re-identification by cross-view multi-level dictionary learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2963–2977, Dec. 2018.
- [23] M. M. Kalayeh, E. Basaran, M. Gökmen, M. E. Kamasak, and M. Shah, "Human semantic parsing for person re-identification," in

- Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 1062–1071.
- [24] L. Zhao, X. Li, Y. Zhuang, and J. Wang, “Deeply-learned part-aligned representations for person re-identification,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 3219–3228.
- [25] G. Wang, Y. Yuan, X. Chen, J. Li, and X. Zhou, “Learning discriminative features with multiple granularities for person re-identification,” in *Proc. 26th ACM Int. Conf. Multimedia (MM)*, 2018, pp. 274–282.
- [26] Y. Sun, L. Zheng, Y. Yang, Q. Tian, and S. Wang, “Beyond part models: Person retrieval with refined part pooling (and a strong convolutional baseline),” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 480–496.
- [27] C. Su, J. Li, S. Zhang, J. Xing, W. Gao, and Q. Tian, “Pose-driven deep convolutional model for person re-identification,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 3960–3969.
- [28] P. Wang, C. Ding, Z. Shao, Z. Hong, S. Zhang, and D. Tao, “Quality-aware part models for occluded person re-identification,” *IEEE Trans. Multimedia*, vol. 25, pp. 3154–3165, 2023.
- [29] D. L. Donoho and M. Elad, “Optimally sparse representation in general (nonorthogonal) dictionaries via ℓ_1 minimization,” *Proc. Nat. Acad. Sci. USA*, vol. 100, no. 5, 2003, pp. 2197–2202.
- [30] H. Yu, “Network complexity analysis of multilayer feedforward artificial neural networks,” in *Proc. Appl. Neural Netw. High Assurance Syst.*, 2010, pp. 41–55.
- [31] J. Gorski, F. Pfeuffer, and K. Klamroth, “Biconvex sets and optimization with biconvex functions: a survey and extensions,” *Math. Methods Operations Res.*, vol. 66, no. 3, pp. 373–407, 2007.
- [32] J. Zhuo, Z. Chen, J. Lai, and G. Wang, “Occluded person re-identification,” in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, 2018, pp. 1–6.
- [33] L. He, Y. Wang, W. Liu, H. Zhao, Z. Sun, and J. Feng, “Foreground-aware pyramid reconstruction for alignment-free occluded person re-identification,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 8450–8459.
- [34] G. Wang et al., “High-order information matters: Learning relation and topology for occluded person re-identification,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 6449–6458.
- [35] S. He, H. Luo, P. Wang, F. Wang, H. Li, and W. Jiang, “TransReID: Transformer-based object re-identification,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 15 013–15 022.
- [36] V. Somers, C. De Vleeschouwer, and A. Alahi, “Body part-based representation learning for occluded person re-identification,” in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2023, pp. 1613–1623.
- [37] S. He et al., “Region generation and assessment network for occluded person re-identification,” *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 120–132, 2024.
- [38] Z. Gao, L. Gao, H. Zhang, Z. Cheng, R. Hong, and S. Chen, “DCR: A unified framework for holistic/partial person ReID,” *IEEE Trans. Multimedia*, vol. 23, pp. 3332–3345, 2021.



Arindam Sikdar (Member, IEEE) received the M.Tech. degree in electronic and communication engineering from IIT Bhubaneswar, Odisha, India, in 2015, and the Ph.D. degree from Jadavpur University, Kolkata, India, in 2021.

He has been associated in the SPHERE project while working as a Postdoctoral Fellow with the University of Bristol, Bristol, U.K. He is currently working in another Postdoctoral position with Edge Hill University, Ormskirk, U.K. He has published nine papers in international journals and conferences. He has been a regular referee for reputed scientific conferences and journals like ICAPR, CVIP, IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), *Pattern Recognition Letters (PRL)*, IEEE SIGNAL PROCESSING LETTERS (SPL), *Multimedia Tools and Applications (MTAP)*, *Expert Systems With Application (ESWA)*, and *Springer Nature (SN)*. His current research interests include computer vision and pattern recognition, visual surveillance, multimedia analysis, and digital healthcare.



Ananda S. Chowdhury (Senior Member, IEEE) received the Ph.D. degree in computer science from the University of Georgia, Athens, GA, USA, in 2007.

He was a Postdoctoral Fellow with the Department of Radiology and Imaging Sciences, National Institutes of Health, Bethesda, MD, USA, from 2007 to 2008. He is currently a Professor with the Department of Electronics and Telecommunication Engineering, Jadavpur University, Kolkata, India, where he leads the Imaging, Vision, and Pattern Recognition group. He has authored or co-authored more than 90 articles in leading international journals and conferences, in addition to a monograph in the Springer series of *Advances in Computer Vision and Pattern Recognition*. His research interests include computer vision, pattern recognition, biomedical image/signal processing, and multimedia analysis.

Dr. Chowdhury is a Member of the International Association for Pattern Recognition Technical Committee (IAPR TC) on Graph-Based Representations. He currently serves as an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING and as an Area Editor for *Pattern Recognition Letters*. In the past, he was a Senior Area Editor of IEEE SIGNAL PROCESSING LETTERS and an Associate Editor for *Springer Nature Computer Science*. His Erdős Number is two.